

Retail Spatial Intelligence

CCTV Foot-Traffic & Spatial Behavior Report

VERIFIED COMPUTER VISION DATA

Analysis Parameters & Provenance

VIDEO SOURCE:	video_1791353940461_00bab785.mp4	DATE PROCESSED:	Oct 7, 2026, 11:52 AM
RESOLUTION:	1270x720 (960x544)	VIDEO DURATION:	111.05s (1454 frames @ 13.09 fps)
ZONE CONFIG:	Custom Retail Zones	ANALYSIS QUALITY:	Medium Quality
CV PIPELINE:	YOLOv8 + ByteTrack (Floor Footprints)	REPORT VERSION:	v2.2 (Canonical Single Truth)
TRACKING QUALITY:	Moderate (90 valid tracks)	JOB AUDIT ID:	05b0ce3a-fa90-4bd6

TRACKED VISITORS

90

Unique valid tracks

AVG ZONE DWELL

7.11s

Median: 4.28s

PEAK CONCURRENCY

13

Max simultaneous

TOTAL ZONE VISITS

153

Across 4 zones

Executive Summary

DETERMINISTIC SPATIAL RULES SUMMARY

Video analysis observed 90 unique tracked customer visits across 111.05s of footage. Overall average dwell time was 7.11s with a peak simultaneous store occupancy of 13 people. 4 potential bottleneck area(s) and 0 potential dead-zone area(s) were deterministically identified.

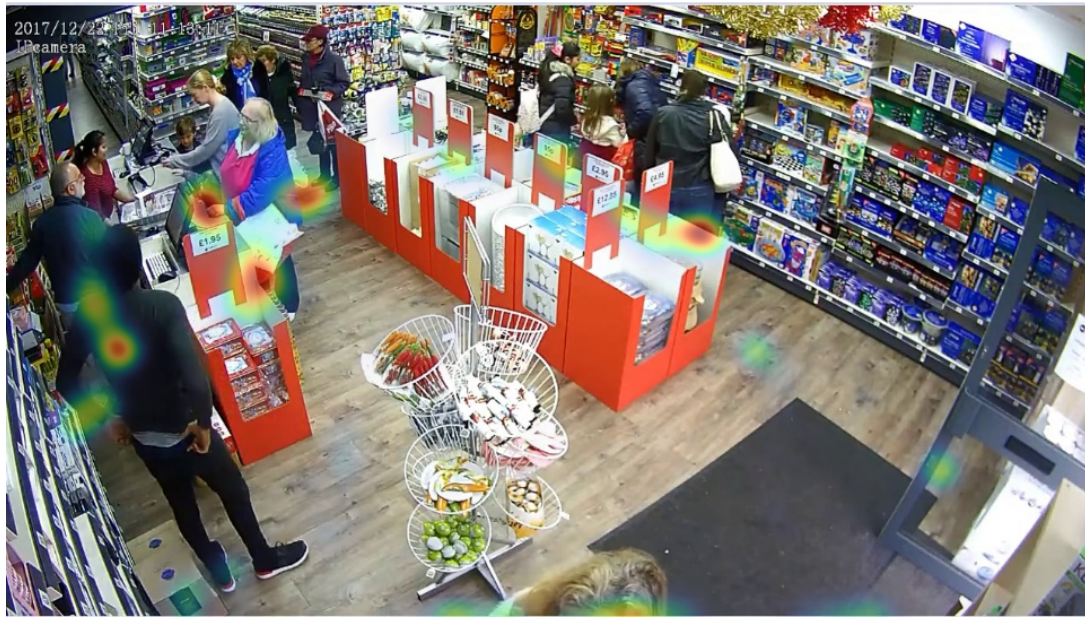
1. Temporal Traffic Dynamics & Concurrency

Primary traffic flow centers around Region 1 (Upper Left). Most common movement pathway: Region 1 (Upper Left). Observed 8 distinct zone-to-zone transitions.

PEAK STORE OCCUPANCY:	13 concurrent tracked individuals
AVERAGE OCCUPANCY:	8.26 people per analyzed interval
IMAGE-SPACE SPEED:	29.4 px/s (Relative pedestrian velocity)

2. 2D Gaussian Kernel Spatial Heatmap

Accumulated ground foot-points smoothed via 2D Gaussian density kernel and rendered onto the representative camera frame:



Heatmap represents observed customer foot-point activity during the analyzed period. Red indicates peak dwell density.

3. Spatial Zone Performance Analysis (Normalized)

Comprehensive empirical measurements per spatial zone. Normalized area metrics prevent large zones from appearing artificially dominant.

Zone Name	Visits	Unique	Avg Dwell	Max Dwell	Peak	Area (%)	Density	Vis/Min	Share
Region 1 (Upper Left)	78	81	6.36s	28.11s	8	25%	1975.38	43.76	51%
Region 2 (Upper Right)	20	14	8.52s	58.04s	3	25%	651.57	7.56	13.1%
Region 3 (Lower Left)	32	36	8.1s	73.02s	5	25%	1179.43	19.45	20.9%
Region 4 (Lower Right)	23	22	7.04s	25.05s	6	25%	782.97	11.89	15%

4. Customer Flow & Traversal Pathways

Zone-to-zone transitions mined from consecutive spatial crossings across verified customer paths:

- DETECTED DIRECT ZONE TRANSITIONS:
- Region 3 (Lower Left) !' Region 1 (Upper Left) : 26 visitor(s) (29.9%)
 - Region 1 (Upper Left) !' Region 3 (Lower Left) : 25 visitor(s) (28.7%)
 - Region 1 (Upper Left) !' Region 2 (Upper Right) : 11 visitor(s) (12.6%)
 - Region 2 (Upper Right) !' Region 1 (Upper Left) : 9 visitor(s) (10.3%)
 - Region 4 (Lower Right) !' Region 3 (Lower Left) : 5 visitor(s) (5.7%)

Frequent Customer Journey Sequences:

1. Region 1 (Upper Left)
Observed in 38 customer(s) (42.2% of journeys) - Avg Duration: 8.9s
2. Region 4 (Lower Right)
Observed in 9 customer(s) (10% of journeys) - Avg Duration: 7.7s
3. Region 3 (Lower Left)
Observed in 9 customer(s) (10% of journeys) - Avg Duration: 6.9s
4. Region 2 (Upper Right)
Observed in 8 customer(s) (8.9% of journeys) - Avg Duration: 14.7s

5. Forensic Bottleneck & Dead-Zone Intelligence

POTENTIAL BOTTLENECK ANALYSIS:

- **Region 1 (Upper Left) (Bottleneck Score: 77.6/100 - Confidence: High)**
 - Evidence: 62% relative store peak occupancy (8 simultaneous customer tracks)
 - Evidence: Movement slowdown: 21.8 px/s (26% slower than store median 29.4 px/s)
 - Evidence: Elevated spatial traffic density: 62.02 visitors per 100k sq px
- **Region 3 (Lower Left) (Bottleneck Score: 57.1/100 - Confidence: High)**
 - Evidence: 38% relative store peak occupancy (5 simultaneous customer tracks)
 - Evidence: Prolonged dwell duration: 8.1s exceeds store average (7.11s)
 - Evidence: Unconstrained movement speed: 35.3 px/s indicates fluid traversal
 - Evidence: Elevated spatial traffic density: 27.55 visitors per 100k sq px
- **Region 4 (Lower Right) (Bottleneck Score: 55.1/100 - Confidence: High)**
 - Evidence: 46% relative store peak occupancy (6 simultaneous customer tracks)
 - Evidence: Unconstrained movement speed: 47.4 px/s indicates fluid traversal
- **Region 2 (Upper Right) (Bottleneck Score: 51.8/100 - Confidence: High)**
 - Evidence: 23% relative store peak occupancy (3 simultaneous customer tracks)
 - Evidence: Prolonged dwell duration: 8.52s exceeds store average (7.11s)
 - Evidence: Movement slowdown: 20.4 px/s (31% slower than store median 29.4 px/s)

POTENTIAL DEAD-ZONE ANALYSIS:

- Healthy foot-traffic distribution observed across all monitored zones.

HIGH-ACTIVITY DESTINATION ZONES:

- Region 1 (Upper Left): 78 visits, 6.36s avg dwell, peak concurrent: 8 (High Activity Corridor)
- Region 3 (Lower Left): 32 visits, 8.1s avg dwell, peak concurrent: 5 (High Activity Corridor)

6. Strategic Recommendations & Priority Action Plan

[Congestion Alleviation] Region 1 (Upper Left)

Evidence: 62% relative store peak occupancy (8 simultaneous customer tracks); Movement slowdown: 21.8 px/s (26% slower than store median 29.4 px/s); Elevated spatial traffic density: 62.02 visitors per 100k sq px
Action: Widen primary aisle clearance or reposition high-interaction displays in Region 1 (Upper Left) to alleviate observed transit bottlenecks.

Prioritized Action Plan:

- [High Priority]:** Audit aisle clearance in Region 1 (Upper Left), Region 3 (Lower Left), Region 4 (Lower Right), Region 2 (Upper Right) to eliminate movement slowdowns.
- [Medium Priority]:** Compare peak occupancy periods with staff scheduling.
- [Low Priority]:** Conduct subsequent video analysis after implementing spatial adjustments to evaluate before/after metrics.

7. Analytics Validation & Ground-Truth Integrity

DETECTION VALIDATION:
5995 raw detections | 0.608 avg conf | 4.13 det/frame

TRACKING VALIDATION:
90 valid tracks (31 transient filtered) | Max duration: 89.06s | ~44 ID switch candidate(s)

Epistemological Scope: What The System Actually Knows

THE SYSTEM CAN DIRECTLY OBSERVE:

- Spatial pedestrian movement & footpaths
- Zone entry/exit timestamps & dwell duration
- Concurrent physical occupancy & density
- Sequential zone transition flow & friction

THE SYSTEM CANNOT DIRECTLY OBSERVE:

- Customer identity or demographic PII
- Product purchase decisions or basket value
- Revenue or sales conversion (requires POS)
- Real-world metric distance without 3D calibration

Scientific Methodology & Limitations

- Single-Camera Monocular Perspective: Field of view is restricted to camera mounting angle. Speeds are reported in image-space pixels/second, not calibrated metric coordinates.
- Visual Occlusion: Dense clusters or shelving can temporarily occlude foot points. The platform uses a tolerance buffer, but prolonged occlusions may fragment tracks.
- Zero PII Policy: The system operates with transient anonymous IDs only. No face recognition, demographic profiling, or biometric storage is performed.
- Automatic Regions: Partitioned geometrically across the camera field of view; they represent spatial camera sectors, not semantic store departments.

AUTHENTICATION & VERIFICATION STAMP

Generated deterministically by Retail Spatial Intelligence Platform Engine v2.2. All metrics directly traceable to video frame detections. Job ID: 05b0ce3a-fa90-4bd6-b71c-0fd17b440b0c

